



-  **Real-time Monitoring**
Live data from sensors
-  **Smart Irrigation**
Automated & efficient
-  **Water Saving**
Use only what's needed
-  **Remote Control**
Anywhere, anytime

— **IoT Based** —
Smart Irrigation System
Smarter Farming | Better Yields | Sustainable Future

USER MANUAL

Technology for a Greener Tomorrow



EFFICIENT
Use water wisely



INTELLIGENT
Data-driven decisions



RELIABLE
Built for the field



SUSTAINABLE
For a better future

Welcome to the AquaSol Smart Irrigation System User Manual.

Thank you for choosing us!

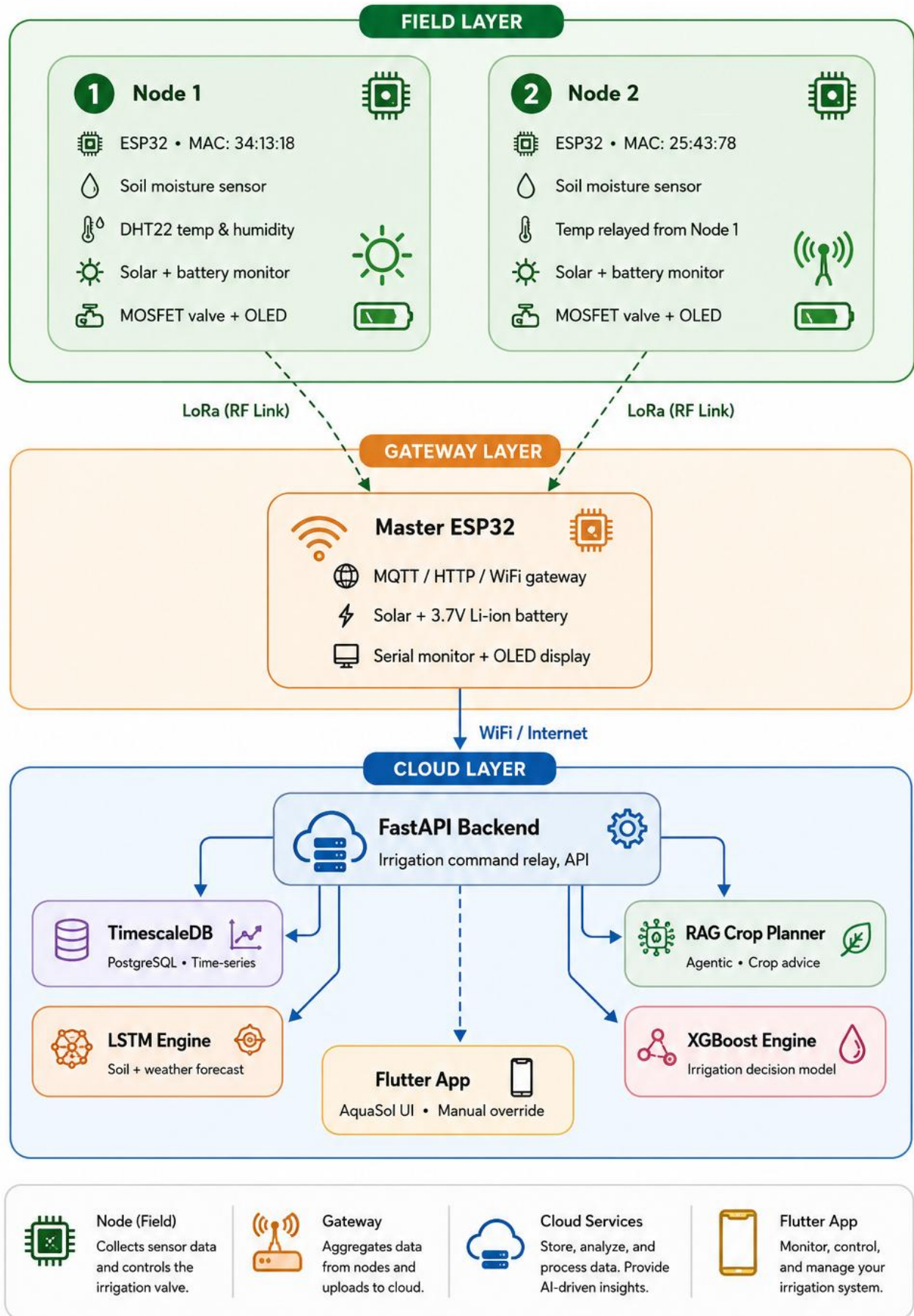
This document provides a complete guide for understanding, operating, and maintaining the Smart Irrigation System developed using ESP32 microcontrollers, LoRa wireless communication, solar-powered sensor nodes, and AI-based irrigation control. The system is designed to improve agricultural efficiency by automating irrigation based on real-time environmental and soil conditions.

The project combines Internet of Things (IoT), wireless communication, renewable energy, and intelligent decision-making technologies to create a reliable and energy-efficient irrigation solution. The system continuously monitors parameters such as soil moisture, temperature, humidity, rain conditions, water flow, battery status, and solar efficiency to optimize water usage and support sustainable farming practices.

This manual is intended for operators, developers, technicians, and users who will install, configure, monitor, or maintain the system. It includes detailed information about system architecture, hardware components, software workflow, communication protocols, operating procedures, troubleshooting methods, and safety features.

By following this manual, users can effectively operate the Smart Irrigation System and understand its functionality, reliability features, and intelligent irrigation capabilities.

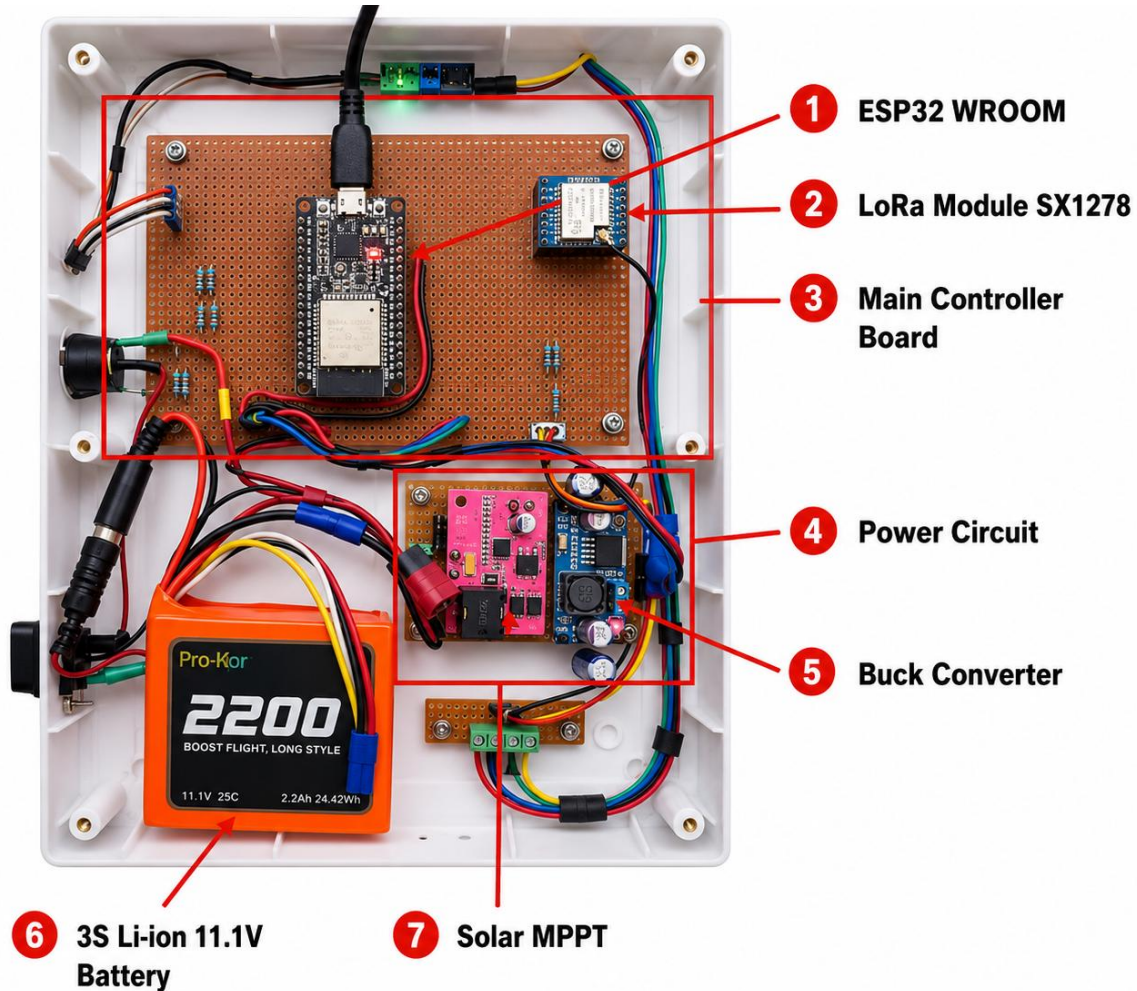
SYSTEM OVERVIEW



HARDWARE OVERVIEW

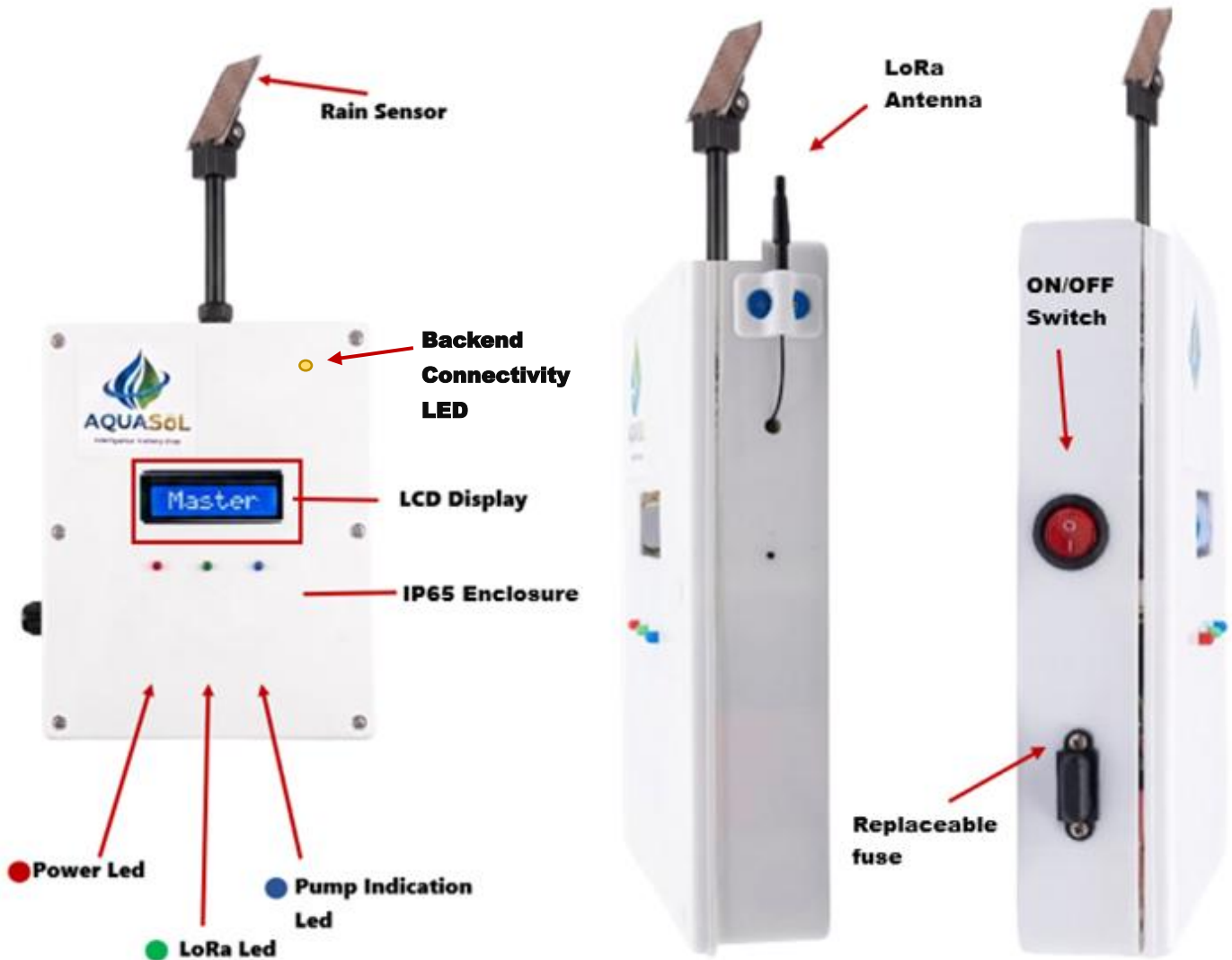
MASTER :

Internal structure



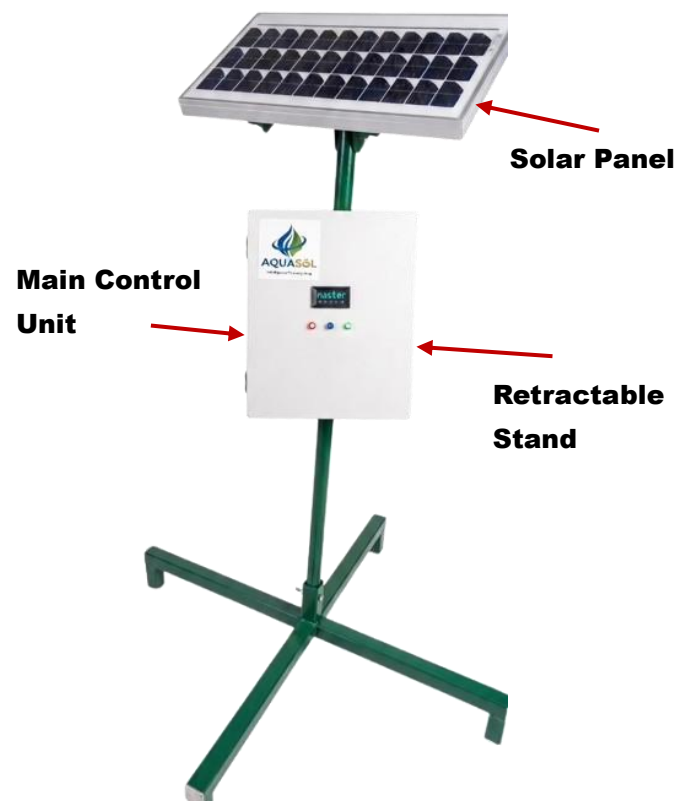
Component	Function & Description
ESP32 Microcontroller	Main control unit. Processes data and manages logic.
LoRa SX1278 Module	Wireless communication module for long-range data.
Rain Sensor	Detects rain conditions to trigger automated logic.
Flow Sensor	Measures water flow and usage for tracking.
I2C LCD Display	Displays system status and sensor readings locally.
Battery Monitoring Circuit	Measures battery percentage and voltage levels.
Solar Voltage Monitoring Circuit	Measures solar efficiency and panel performance.
LED Indicators	Shows power, pump, and LoRa status via visual cues.

Design



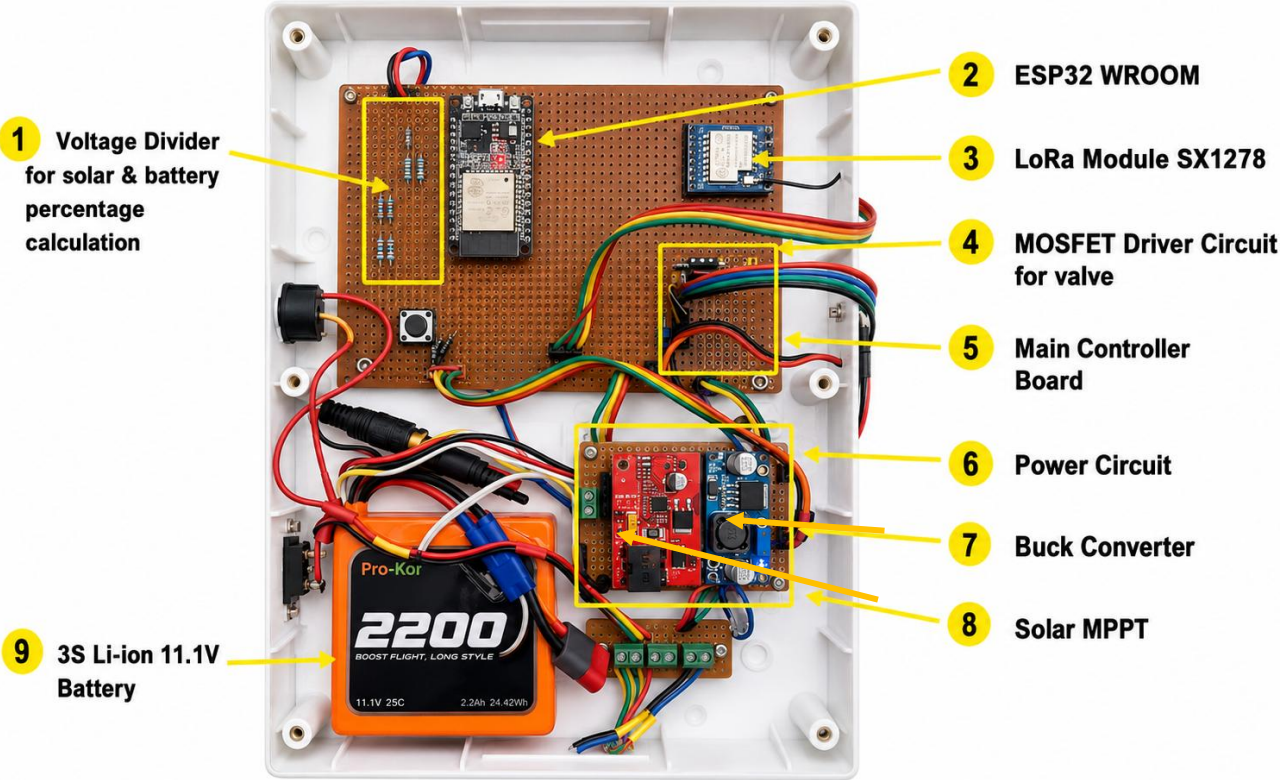
Complete Node

A solar panel is integrated into the system to provide renewable energy for continuous operation in outdoor agricultural environments. The solar panel charges the battery during daytime, allowing the system to operate efficiently even in remote areas without a constant power supply.



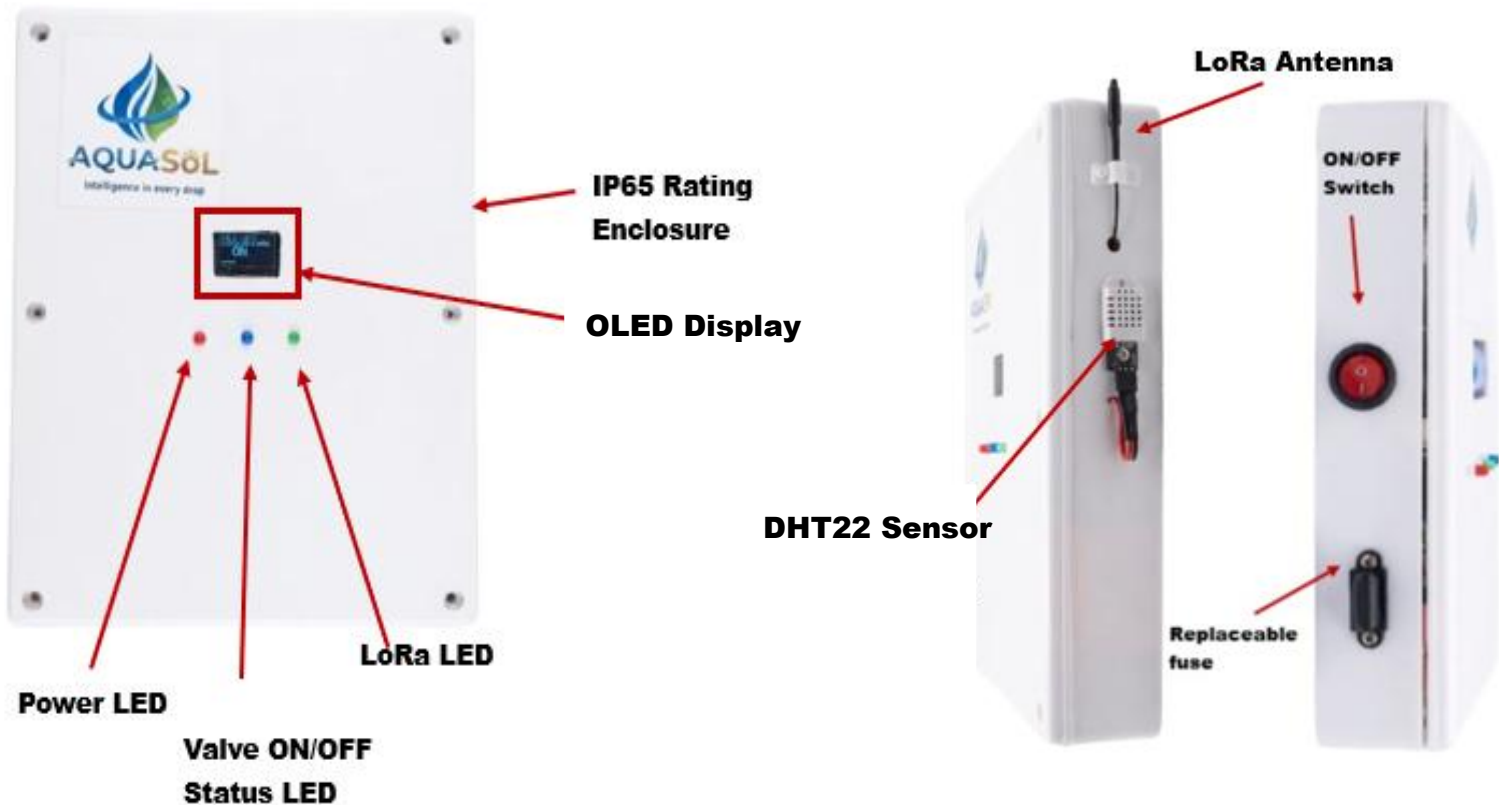
Slave Node:

Internal structure



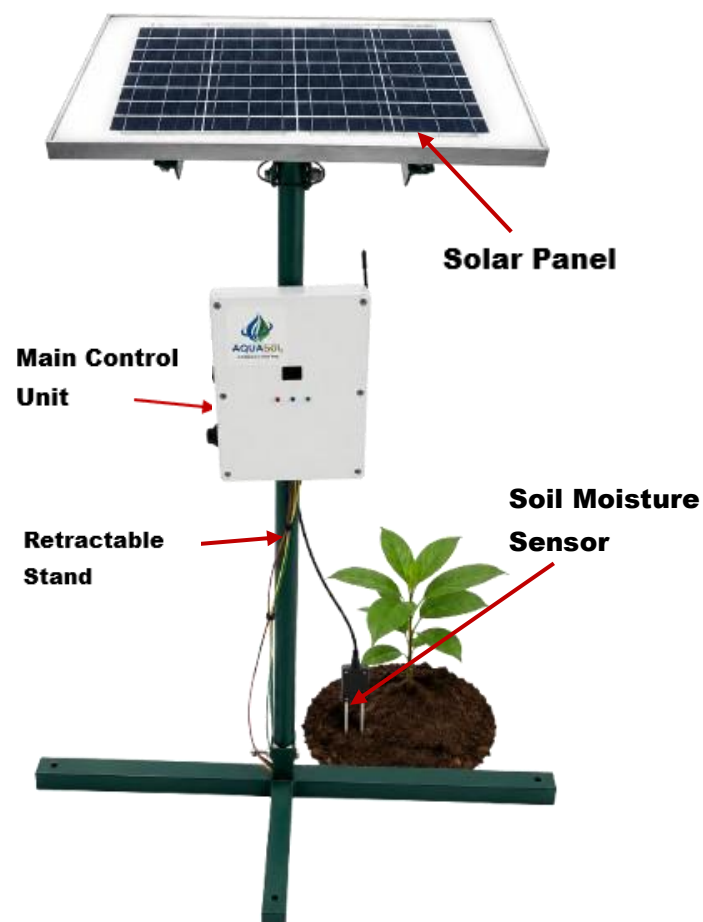
Component	Function & Description
ESP32 Microcontroller	Main control unit. Processes data and manages logic.
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Solar Voltage Monitoring Circuit	Measures solar efficiency and panel performance.
LED Indicators	Shows power, pump, and LoRa status via visual cues.

Design



Complete Node

The Sensor Node is mounted on a durable metal stand designed to securely support the solar panel, control box, sensors, and wiring system. A solar panel is fixed at the top of the structure to provide renewable power for continuous outdoor operation. The generated solar energy charges the battery system, allowing the node to function efficiently even in remote agricultural areas without direct electrical supply. The node enclosure contains the ESP32, LoRa module, power management circuits, and valve control system, while the soil moisture sensor is placed directly into the soil near the plant for accurate real-time monitoring.



HARDWARE INSTALLATION GUIDE

Smart Irrigation System



This guide explains the step-by-step process for installing all hardware components of the Smart Irrigation System. Follow all instructions carefully for safe and reliable operation.

SAFETY PRECAUTIONS

- Ensure power is OFF before making any connections.
- Use proper tools and insulated wires.
- Install the system in a dry and safe environment.
- Ensure solar panel is securely mounted.
- Avoid exposing electronics to water.
- Check all connections before powering ON.

TOOLS REQUIRED

- Screwdriver Set
- Drill Machine
- Spanner / Wrench
- Wire Cutter & Stripper
- Insulation Tape
- Multimeter



1. MASTER NODE INSTALLATION

1 Mount the Stand



Fix the stand firmly on a stable and flat surface using bolts or screws.

2 Install Solar Panel



Fix the solar panel on the top bracket and tighten the screws.

3 Mount Control Box



Attach the master control box to the stand and secure it properly.

4 Connect Power



Connect solar panel wires to the charge controller and battery.

5 Install Sensors



Install rain sensor in open area and flow sensor in the main water line.

6 Power ON & Check



Power ON the system and verify LCD display, LEDs and LoRa status.

★ Ensure the master node is installed at an elevated position for better LoRa range and WiFi connectivity.

2. SENSOR NODE (NODE 1 / NODE 2) INSTALLATION

1 Mount the Stand



Fix the node stand firmly in the field near the crop area.

2 Install Solar Panel



Attach the solar panel on top and tighten the mounting screws.

3 Mount Node Box



Fix the node enclosure on the stand and secure it properly.

4 Connect Power



Connect solar panel to charger and battery. Check all connections.

5 Install Soil Sensor



Insert the soil moisture sensor in the root zone at proper depth.

6 Connect Valve



Connect solenoid valve to the node using MOSFET driver output.

🌿 Repeat the same process for Node 1 and Node 2. Place nodes in representative locations of the field.

3. FINAL CHECKS



Check Power

Verify battery voltage and solar charging status on all nodes.



Check LoRa

Ensure LoRa communication between nodes and master.



Check WiFi

Verify WiFi connection of master node with backend server.



Check Actuation

Test valve ON/OFF operation from backend or manually.



Check Displays

Verify OLED (nodes) and LCD (master) are showing correct information.



System Ready

After all checks, system is ready for operation.

🛡️ **IMPORTANT:** Ensure all connections are tight and weatherproof. Use cable ties and protective conduits for outdoor installation.

SOFTWARE OVERVIEW

HOME SCREEN OVERVIEW

The Home screen gives you a quick sanpshot of your farm's health and key updates.

5:082 devices

100%39%

Dashboard

Greeting, vijay

300°C, Clouds

Farm Health

Optimal

90/100

Today's Water

0.0 L

Active Zones

8/8 On

Ai Decisions

0

Alerts

0 New

Ai Advisory

Welcome to AquaSoll Once your nodes are paired, Solu will

Home

Farm

Control

Data

Solu

1 Side Menu

Tap to access navigation options.

2 Language

Change the app language.

3 Notifications

View alerts and important updates.

4 Profile

Access your profile and account settings.

5 Farm Health Score

Overall health of your farm.
Higher score means healthier farm.

6 Today's Water

Shows the total water used today.

7 Active Zones

Displays how many zones are active out of total zones.

8 AI Decisions

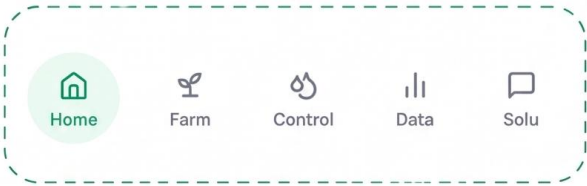
Total number of AI recommendations available.

9 Alerts

View the number of new alerts generated.

10 AI Advisory

Personalized tips and suggestions from AI to improve your farm.



BOTTOM NAVIGATION

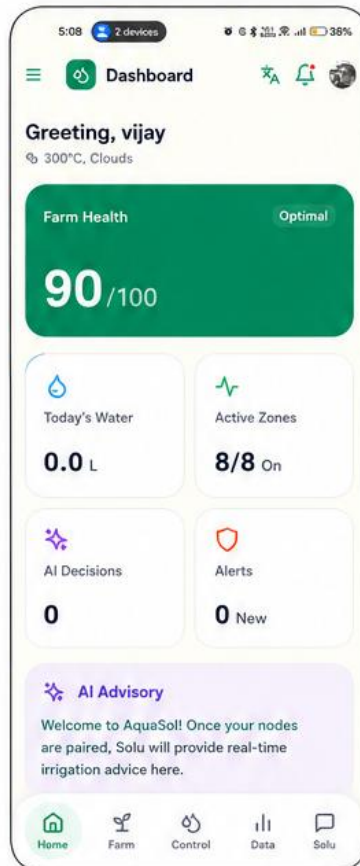
- Home** Go to the Home dashboard.
- Farm** Manage your farm zones, and infrastructure.
- Control** Manually control inigation and devices.
- Data** View insights, charts, and historical data.
- Solu** Chat with Solu – your AI farming assistant.

Explore the main sections of AquaSol app and what you can do in each.

1. HOME

Get a quick overview of your farm's health and key updates.

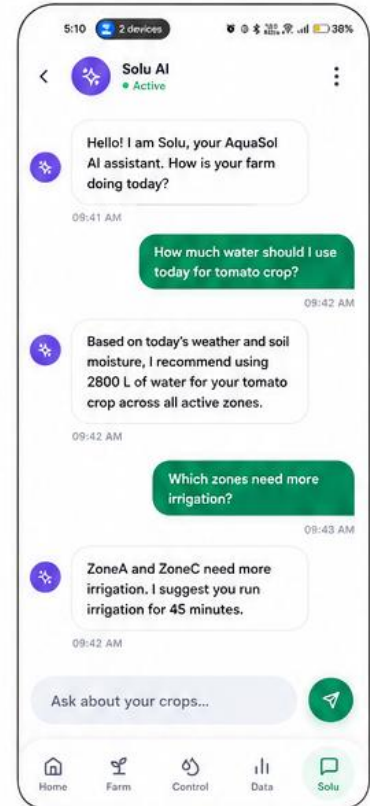
- View Farm Health Score
- Check Today's Water Usage
- See Active Zones
- Get AI Decisions & Alerts
- Receive AI Advisory updates



2. AI (SOLU AI)

Chat with Solu AI for smart farming advice and support.

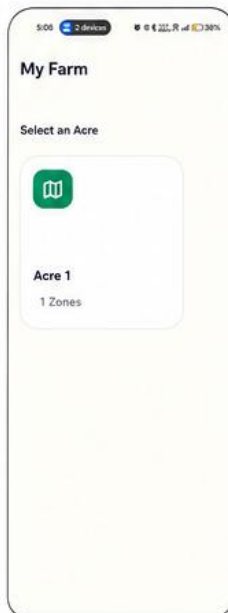
- Get instant answers
- Receive irrigation recommendations
- Ask about crops, zones, and devices
- Your smart farming assistant



3. FARM

Manage your farm structure and view all acres.

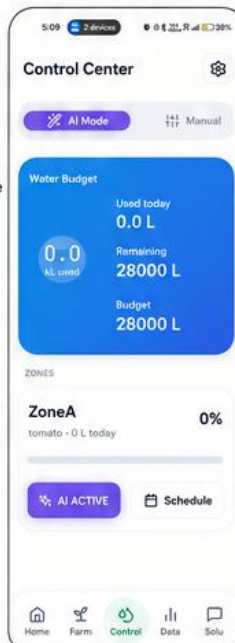
- View all your acres
- See zones under each acre
- Easily switch between acres



4. CONTROL CENTER

Control irrigation with AI or manually and manage water budget.

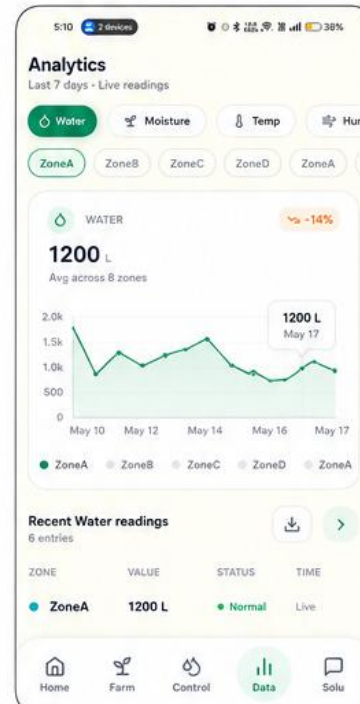
- Choose AI Mode for automatic irrigation
- Switch to Manual for full control
- Monitor water used, remaining & budget
- Manage zones & irrigation schedule



5. DATA (ANALYTICS)

View live sensor data and analytics to make better decisions.

- View data for Water, Moisture, Temp & Humidity
- Filter by zones
- See graphs of last 7 days
- Check recent sensor readings in real-time



QUICK NAVIGATION

Use the bottom navigation bar to quickly switch between sections.



SOFTWARE INSTALLATION GUIDE

Follow these simple steps to set up your system and get started.

1 TELL US ABOUT YOUR FARM

5:03 2 devices 40%

← Skip

Step 1 of 5

Tell us about your farm
A few quick details to set things up

FARM NAME
vijay's Farm

OWNER NAME
vijay

STATE Karnataka **CITY/VILLAGE** Enter City

Auto-Detect Farm GPS Location

Continue

Enter your farm details and location.

2 WHAT ARE YOU GROWING?

5:03 2 devices 40%

← Skip

Step 2 of 5

What are you growing?
Pick your primary crop and soil type

PRIMARY CROP
Search for a crop...

WEEKS SINCE SOWING
0 1 2 3 4 5 6 7 8

Continue

Select your primary crop and weeks since sowing.

3 INFRASTRUCTURE CONFIG

5:03 2 devices 40%

← Skip

Step 3 of 5

Infrastructure Config
Define your farm's irrigation hierarchy

TOTAL ACRES
2

NUMBER OF ZONES PER ACRE
4

NODES PER ZONE
3

WATER SOURCE
Borewell Canal Tank Mixed

You are setting up 8 total zones with 3 nodes each.

Continue

Configure your farm's irrigation structure and water source.

4 NAME YOUR HARDWARE

5:07 2 devices 39%

← Skip

Step 4 of 5

Name Your Hardware
Give friendly names to your gateway and field nodes.

MASTER GATEWAY NAME
Master Gateway

ACRE NAME
Acre 1

ZoneA DISPLAY NAME
ZoneA

DEVICE NAME
NodeA1

DEVICE NAME
NodeA2

DEVICE NAME
NodeA3

Continue

Provide names to your gateway, acre, zones, and devices.

5 HARDWARE DISCOVERY

5:07 2 devices 39%

← Skip

Step 5 of 5

Hardware Discovery
Select your discovered hardware from the list below.

Master Gateway
Click to Enter Pairing Code

Acre 1

ZoneA - NodeA1
Click to Enter Pairing Code

ZoneA - NodeA2
Click to Enter Pairing Code

ZoneA - NodeA3
Click to Enter Pairing Code

ZoneB - NodeB1
Click to Enter Pairing Code

ZoneB - NodeB2
Click to Enter Pairing Code

ZoneB - NodeB3
Click to Enter Pairing Code

Go to Dashboard

Discover and select your hardware from the list.

YOU'RE ALL SET!

Your software is installed and your system is ready to use.

Tap "Go to Dashboard" to start.

MAINTENANCE GUIDE

Regular maintenance keeps your system accurate, reliable, and long-lasting.

Follow these important steps for optimal performance.

01

SENSOR CLEANING

Keep sensors clean for accurate readings.

- Wipe sensors gently with a soft, damp cloth.
- Avoid using harsh chemicals or high-pressure water.



02

BATTERY INSPECTION

Ensure batteries are healthy and functioning.

- Check battery level in the app regularly.
- Replace batteries if level is low or if performance issues occur.



03

SOLAR PANEL MAINTENANCE

Clean solar panels for maximum efficiency.

- Wipe the panel surface with a soft, damp cloth.
- Remove dust, leaves, and debris regularly.



04

FIRMWARE UPDATES

Keep your system up to date for the best performance.

- Check for firmware updates in the app.
- Install updates when available to get new features and fixes.



05

CONNECTION CHECKING

Ensure stable connection for uninterrupted data flow.

- Check device connectivity in the app.
- Ensure gateway is within range and has power.



TIP

Perform these maintenance tasks regularly to extend the life of your devices and ensure accurate farm monitoring.



RECOMMENDED FREQUENCY

Sensor Cleaning: Every 15 days
Battery Inspection: Every 30 days
Solar Panel Cleaning: Every 30 days
Firmware Check: Monthly
Connection Check: Weekly



TROUBLESHOOTING GUIDE

Find common issues, their possible causes, and easy solutions.

! PROBLEM	🔍 CAUSE	✅ SOLUTION
 LoRa communication failure Devices are offline or not communicating.	• Gateway is out of range. • Antenna is loose or damaged. • Obstruction (walls, metal objects, trees). • Low battery in node.	✅ Check if the gateway is within range of all nodes. ✅ Ensure antenna is tightly connected and in good condition. ✅ Reposition the gateway to a higher location with a clear line of sight. ✅ Check node battery level and replace or recharge if needed. 
 Sensor malfunction Sensor readings are incorrect or not updating.	• Sensor is dirty or covered in soil. • Sensor cable is loose or damaged. • Sensor is physically damaged. • Incorrect sensor configuration in app.	✅ Clean the soil moisture sensor probes gently. ✅ Check the cable connections and tighten if loose. ✅ Inspect the sensor for any physical damage. ✅ Verify sensor settings in the app and reconfigure if needed. 
 Valve not activating Irrigation valve does not open or close.	• Manual override is ON. • No power to the valve. • Valve wiring is incorrect or loose. • Valve is blocked or damaged.	✅ Turn off manual override in the app. ✅ Check power supply to the valve and controller. ✅ Inspect valve wiring and connections. ✅ Clean the valve and remove any blockage or debris. 
 WiFi disconnect Gateway or controller is offline.	• Weak WiFi signal. • Incorrect WiFi password. • Router reboot or power issue. • Internet service down.	✅ Move the gateway closer to the WiFi router. ✅ Verify WiFi password and reconnect. ✅ Restart the router and gateway. ✅ Check your internet connection. 
 OLED blank screen Display is not showing anything.	• Display power off. • Loose or disconnected ribbon cable. • Incorrect brightness setting. • Display hardware failure.	✅ Ensure the device is powered ON. ✅ Check and secure the ribbon cable connection. ✅ Increase brightness in display settings. ✅ If the issue persists, contact support for display replacement. 



Thank You!

Thank you for using the AquaSol Smart Irrigation System.



CONTACT INFORMATION

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PROJECT TEAM



HARDWARE TEAM

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